

Speculative Self-Distillation enables Efficient Knowledge Internalization

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TL;DR

Problem. Self-distillation teaches a model a document by training it to match itself with the document in context. **on-policy** rollouts are accurate but spend most tokens where the teacher adds little.

Method. **SSD**: the student writes the rollout; the teacher writes only **inflection tokens**, where the two models disagree by more than τ .

Result. **SSD** matches **on-policy** accuracy with **45% fewer supervised tokens** on three tasks and **forgets less**.

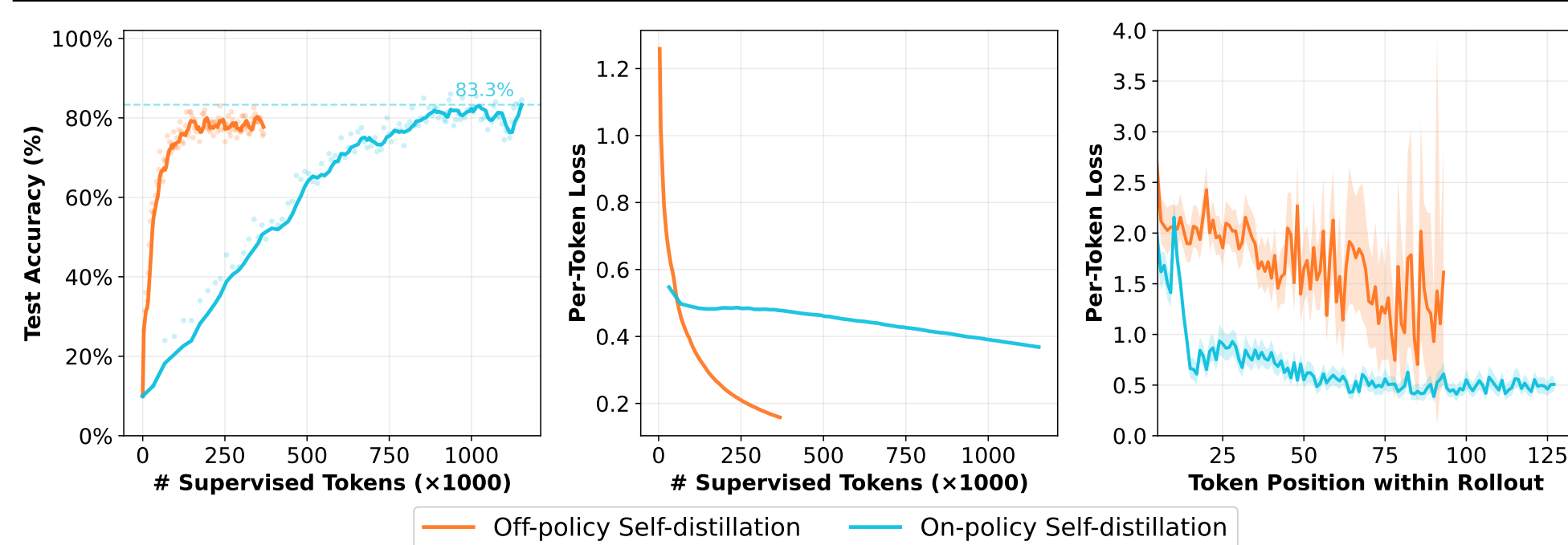
Self-Distillation

- **Teacher** $\pi_T(\cdot | x, C)$: the model with privileged context C (a document). **Student** $\pi_\theta(\cdot | x)$: the same model without C .

$$\mathcal{L}(\theta) = \mathbb{E}_{y \sim \pi_{\text{gen}}} \left[\frac{1}{|y|} \sum_t D(\pi_T(\cdot | x, C, y_{<t}) \| \pi_\theta(\cdot | x, y_{<t})) \right]$$

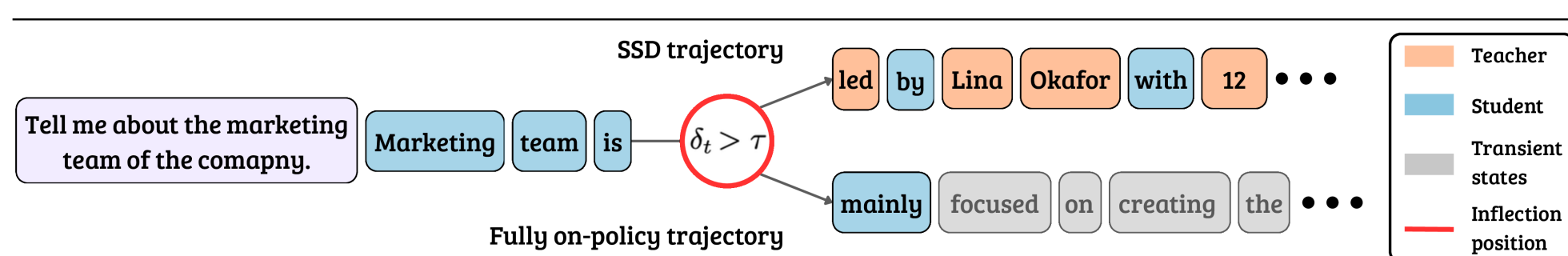
- Methods differ only in the rollout policy π_{gen} ; **SSD** mixes the two per token:
 - **off-policy** ($\pi_{\text{gen}} = \pi_T$): strong signal, but trains on states the student never visits.
 - **on-policy** ($\pi_{\text{gen}} = \pi_\theta$, SDFT): no train-test mismatch, but slow.

Why On-Policy Is Slow



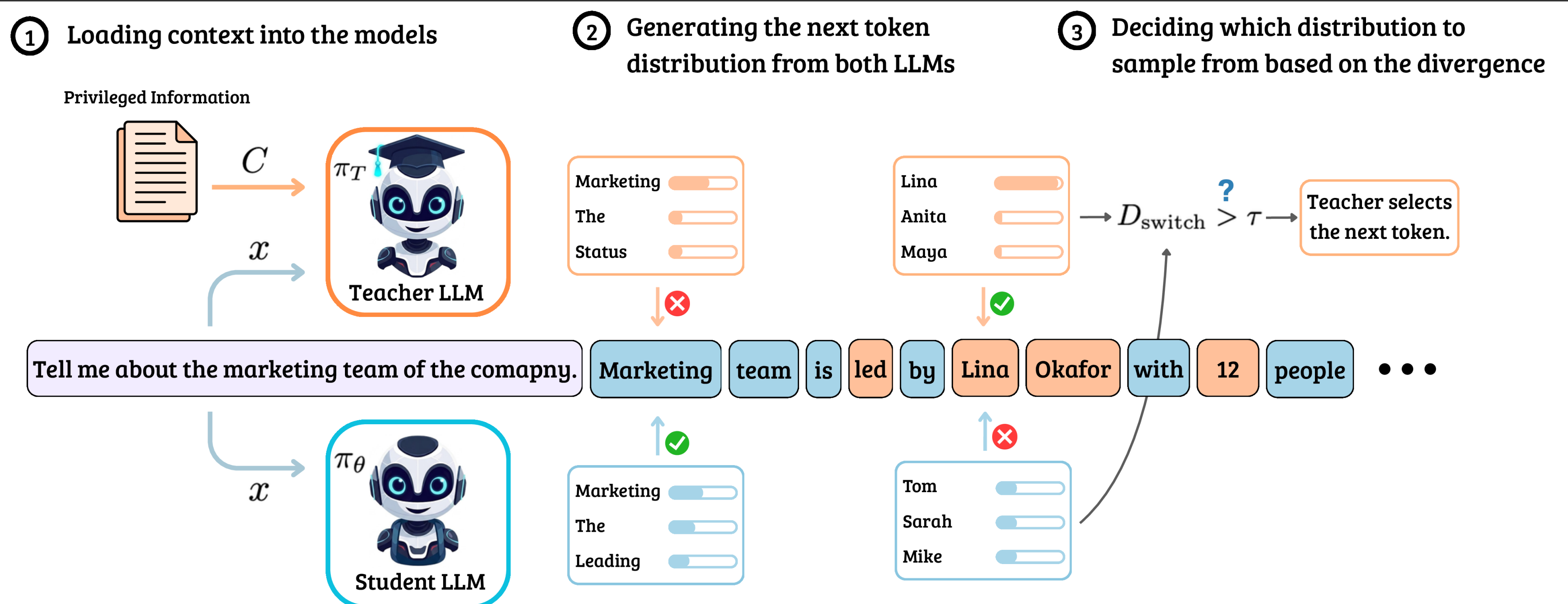
- **Signal** = per-token loss: how much the student can still learn from a token.
- **off-policy** learns $\sim 5\times$ faster but plateaus lower.
- Along **on-policy** rollouts the per-token signal **collapses after a few tokens**.

Inflection Tokens



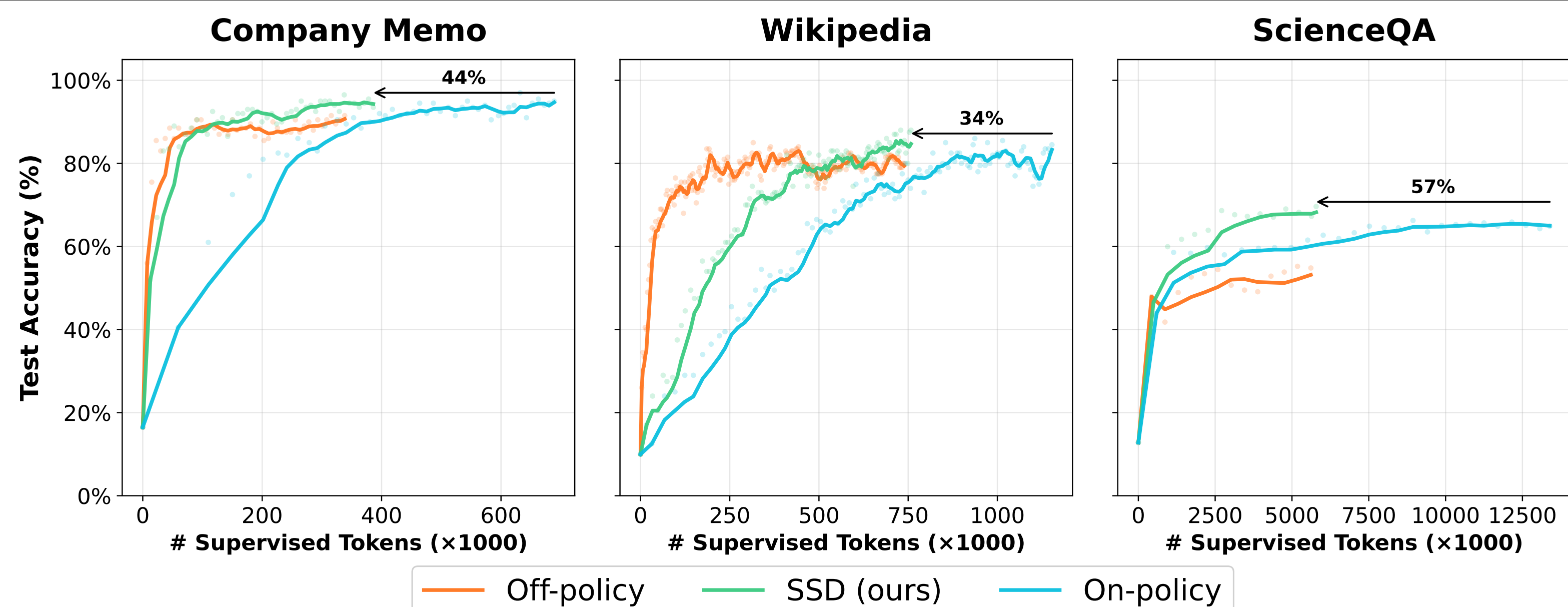
- Where the document changes the next token ($\delta_t > \tau$), **SSD** lets the teacher write.
- **on-policy** keeps the student's token and continues into **transient states**: text the student stops producing after the update, so supervision on it is wasted.

Speculative Self-Distillation (SSD)

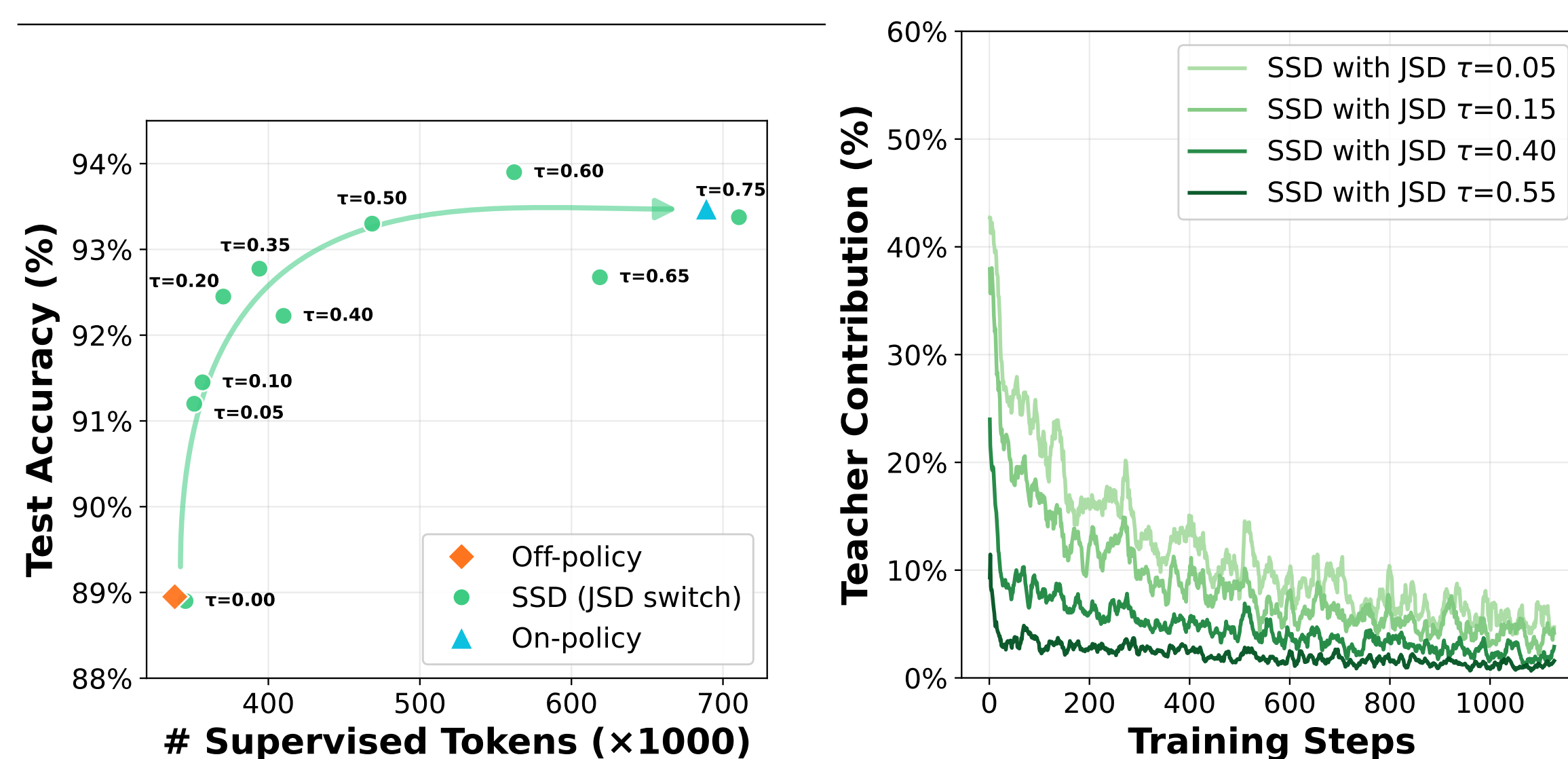


- Student generates by default.
- If $D_{\text{switch}}(\pi_\theta \| \pi_T) > \tau$, the teacher picks the token.
- Every position is trained with the same loss.
- $\tau \rightarrow 0$ is **off-policy**, $\tau \rightarrow \infty$ is **on-policy**.

SSD Reaches On-Policy Accuracy with 34–57% Fewer Tokens

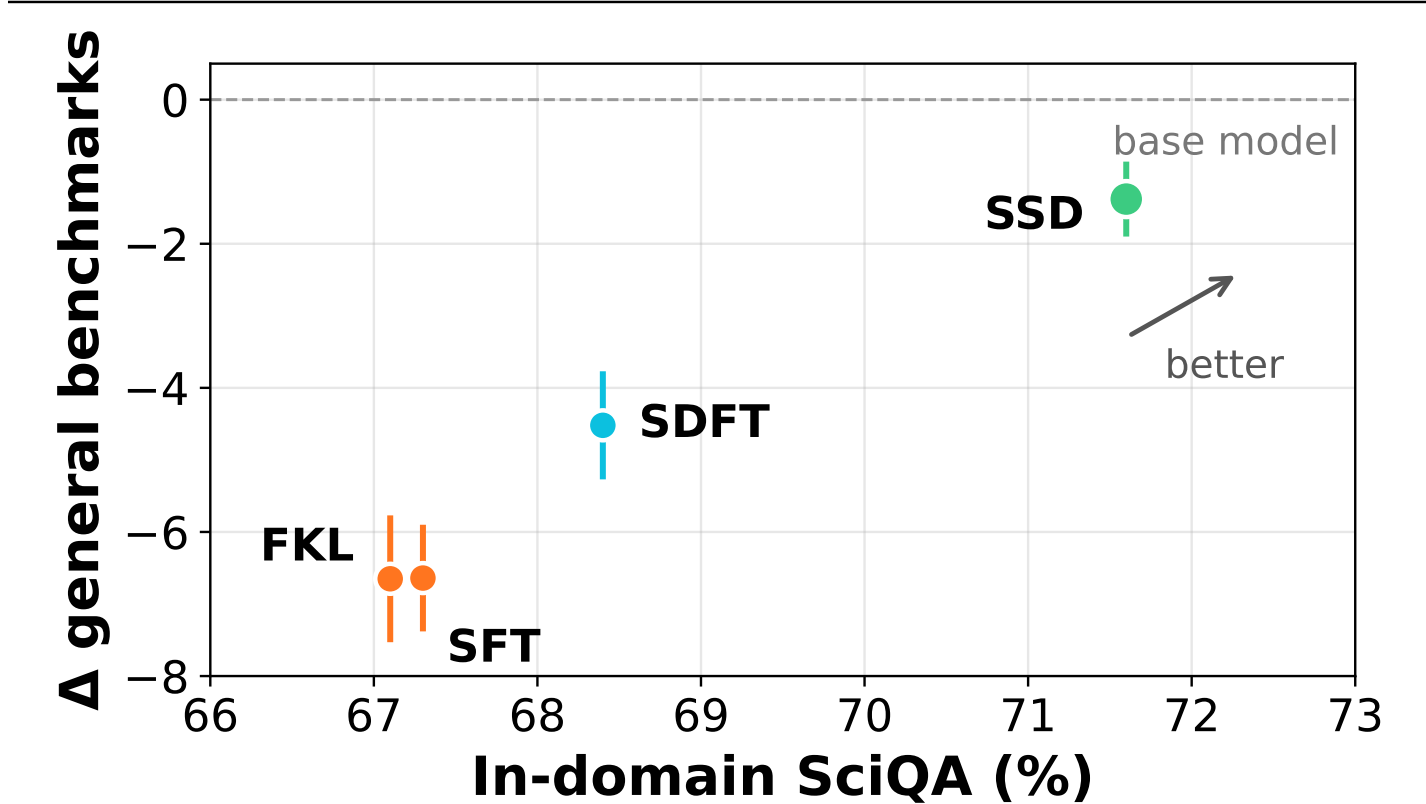


One Knob, and a Built-in Curriculum



- **Left**: small τ is cheap but caps accuracy, like **off-policy**; large τ approaches **on-policy** accuracy and cost. $\tau \in [0.2, 0.5]$ (JSD) sits on the **Pareto frontier**.
- **Right**: the teacher writes up to 40% of tokens early on. As the student learns the document the switch fires less, so training becomes **on-policy without a schedule**.

Less Forgetting



- Qwen2.5-7B trained on SciKnowEval chemistry; y-axis is the mean change on six general benchmarks (MMLU, IFEval, HumanEval, ...).
- **off-policy** SFT/FKL lose ~ 6.6 points, **on-policy** SDFT loses 4.5.
- **SSD**: best SciQA (**71.6**), smallest drop (**-1.4**), 22% fewer tokens than SDFT.